

Atandra Bharati

Deep Learning Research Engineer · 14 from-scratch PyTorch projects across LLMs, latent diffusion, multimodal, agentic ML, long-context attention, and state-space models.
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14 From-scratch PyTorch projects	78% Peak-memory cut – LLaMA-3 pretraining	0.0947 Best loss – Stable Diffusion 1.x from scratch	878 Passing tests – Autonomous ML Research Engineer
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SUMMARY

Self-taught deep learning research engineer with 14 from-scratch PyTorch projects spanning LLMs, latent diffusion, multimodal AI, agentic research, long-context attention, and state-space models. Focused on GPU memory engineering, single-GPU feasibility, paper-faithful reproduction, and agentic orchestration of ML research.

Headline results: a 78% peak-memory cut on LLaMA-3 pretraining (92 GB → 20 GB); a 0.0947 training loss on Stable Diffusion 1.x trained from scratch (860M UNet, 2× RTX 5090); a 2× KV-cache reduction at 128K context in GPT-OSS-Lite via sliding-window/full-attention alternation; and a 15-phase multi-agent ML research platform with 878 passing tests.

EXPERIENCE

ML Engineering Portfolio Nov 2022 – Present · Kolkata, India
Self-directed · Full-time · github.com/atandra2000
Designed, implemented, and trained 14 end-to-end production-grade deep learning systems in raw PyTorch – no high-level wrappers, no pretrained shortcuts where the model itself is the subject. Every project engineered to fit on a single A100 80GB, 2× RTX 5090, or smaller via BF16, Flash-Attn 2, gradient checkpointing, torch.compile, and chunked cross-entropy.
Coverage: LLM pretraining (DeepSeek-V3 reproduction, LLaMA-3, FusionLLM hybrid, GPT-OSS-Lite long-context MoE, Mamba-3 complex64 SSD); latent diffusion (Stable Diffusion 1.x from scratch on 1.3M+ images across 7 phases); multimodal (PaliGemma-style VLM on COCO with zero pretrained weights); generative vision (CycleGAN, β-VAE, DCGAN); video understanding (HRNet pose + ST-GCN); and an agentic ML research platform.

SELECTED PROJECTS Six flagship builds · 2022–2026

Stable Diffusion 1.x From Scratch Latent Diffusion · Generative Vision
Trained a complete Stable Diffusion 1.x model from random initialization on 2× RTX 5090, releasing the 42-epoch checkpoint. 860M-param UNet across a 7-phase curriculum on 1.3M+ images (LAION-Aesthetic → high-aesthetic LAION → DiffusionDB/JourneyDB → VGGFace2 → COCO → consolidation); best loss 0.0947 at epoch 16. Atomic .tmp → os.replace checkpointing, Min-SNR loss, GPU-resident EMA (0.9999), channels_last on Blackwell, DDP/NCCL with no_sync accumulation.
860M UNet · 7 phases · 1.3M+ images 2× RTX 5090 · 42 epochs · best loss 0.0947

LLaMA-3-Lite – 78% Memory Reduction on a Single A100 LLM · Memory Engineering
Built a 515M-param LLaMA-3-style transformer engineered to fit on a single A100 80GB, cutting peak training memory by 78% (92 GB → 20 GB). The cut stacks gradient checkpointing, chunked cross-entropy (logits 50 GB → 0.3 GB), and disk-backed token caching (RAM 112 GB → 1 MB) – enabling 2× batch-size headroom and ~33% throughput gain. GQA (8Q/4KV), RoPE (θ = 500K), SwiGLU, RMSNorm, Flash-Attn 2, BF16, safetensors with full RNG-state restore.
515M params · 8.25B-token run A100 80GB · Chunked CE · BF16 · FA2

Autonomous ML Research Engineer Agentic Platform · Research Infra
A 15-phase multi-agent platform that turns an arXiv paper into evaluated ML experiments end-to-end – paper analysis, repo analysis, experiment planning, code patches, training runs, statistical evaluation, autonomous looping, and research reports. Provider-agnostic LLM layer (Qwen3-Coder, GLM-5.2, Ollama Cloud) with per-agent routing, vector + graph memory, Pydantic v2, self-repair. 878 passing tests across the agent and tool surface.
23 agents · 61 tools · 186 models · 878 tests Qwen3-Coder · GLM-5.2 · Ollama Cloud · Pydantic v2

DeepSeek-V3-Lite – Faithful V3 Reproduction LLM · Frontier Architecture
Faithful from-scratch reimplement of DeepSeek-V3 at 422M params – MLA, aux-loss-free MoE, MTP, end-to-end. MLA with KV low-rank compression, the absorption trick, decoupled RoPE; DeepSeekMoE with aux-loss-free biased-sigmoid balancing; MTP with true absorption at inference. 8.4B-token Chinchilla recipe on a single A100 80GB via SDPA Flash-Attn 2, μP LR scaling, torch.compile(max-autotune); speculative decoding up to 2× throughput via the MTP draft head. Author of the 643-line MLA deep-dive.
422M params · MLA · MoE · MTP A100 80GB · SDPA · torch.compile · μP · 8.4B-token recipe

GPT-OSS-Lite – Long-Context MoE (2× KV-Cache Cut at 128K) LLM · Long-Context Attention
From-scratch reimplementation of OpenAI's GPT-OSS at 502M total / 247M active params. Twelve layers alternate sliding-window (window=128) and full attention; per-head learned attention-sink bias (clamped to [-10, 15] for BF16 SDPA stability); YaRN RoPE (0=100K, scale=32, target=128K); pruned RoPE on global layers; top-2-of-8 routed experts + 1 shared. Verifies a 2× KV-cache reduction at 128K context (1.13 GB vs 2.25 GB pure GQA, BF16). 130 tests across 10 files. Companion ATTENTION_SINKS.md walks through the sink-bias trick.

502M total / 247M active · SWA(128)/Full alt Learned sinks · YaRN 128K · top-2-of-8 MoE · 130 tests

Mamba-3-Lite – Complex64 SSD with Zero Causal Conv LLM · State-Space Model
From-scratch reproduction of Mamba-3 at 404M params, written in pure PyTorch – no mamba-ssm, no causal_conv1d, no custom CUDA, no Triton. Promotes the SSD recurrence into the complex plane (N=64, complex64) so one complex state captures both decay and rotation, halving the state dimension of Mamba-2 at parity capacity. Adds a fully-connected MIMO mixer across SSM heads and removes the causal-conv block. Companion SSD.md derives the chunkwise algorithm.

404M params · complex64 SSD (N=64) · MIMO head mixing A100 80GB · pure PyTorch · zero causal conv

FusionLLM – Hybrid MLA + GDN + MoE + MTP LLM · Hybrid Architecture
24-layer single-GPU pre-training framework fusing four frontier mechanisms in one model: MLA, Gated Delta Net (linear-complexity attention via chunked delta-rule), DeepSeek-style MoE, and Multi-Token Prediction. 415.6M active / 868.6M stored params, 64K vocab, seq_len 4096. Dual-optimizer strategy (NorMuon for 2D matrices, CautiousAdamW for embeddings and norms), WSD schedule, μ P init, logit softcap ± 15.0 , FP32-precise GDN recurrence. 6-stage resumable data pipeline, 55 unit tests.

415.6M active / 868.6M stored NorMuon + CautiousAdamW · WSD + μ P · 8.31B-token recipe

ADDITIONAL PROJECTS Six more · each with verified repo and build log

- *Vision-Language Model (PaliGemma-inspired)*. SigLIP vision encoder + linear projector + Gemma-style GQA decoder · trained end-to-end on COCO 2014 captions with zero pre-trained weights · P100.
- *Face Aging CycleGAN*. Per-layer AdaIN conditioning on a 101-class age embedding, 3-scale PatchGAN + mean-residue age loss, LSGAN + VGG-19 perceptual + L1 identity · 31/50 epochs on IMDB-Wiki (500k faces) · RTX 6000 Ada.
- *Face Generation β -VAE*. 128×128 CelebA, bilinear-upsample decoder (no ConvTranspose), linear KL annealing 0 → 1 over 30 epochs · recon MSE 0.0152 · P100.
- *DCGAN Face Generation*. Faithful 2015 paper reproduction (Radford et al.), 50 epochs on 202k CelebA, D loss converges to $\ln 2 \approx 0.693$ (theoretical Nash equilibrium) · 2× T4 Kaggle.
- *Neural Machine Translation (EN → IT)*. Encoder-decoder Transformer (6+6, d_model = 512), custom SentencePiece BPE, cross-attention visualization · loss 6.17 → 2.28 over 20 epochs on opus_books · P100.
- *Action Recognition (ST-GCN)*. Real-time skeleton-based action recognition with from-scratch HRNet-like pose estimation + two-stream CTR-GCN · 120-class NTU RGB+D 120 pipeline ready, real-time inference targeting 30 FPS on RTX 3090.

TECHNICAL SKILLS Built up across fourteen from-scratch projects · every keyword has shipped code behind it

LLM	LLaMA-3, DeepSeek-V2/V3, GPT-OSS, Mamba-2/3, MLA, GQA, MQA, MoE, MTP, Gated Delta Net, complex64 SSD, sliding-window + full-attention alternation, learned attention sinks, YaRN, Chinchilla scaling, μ P, WSD, NorMuon, CautiousAdamW, AdamW, speculative decoding.
Generative Vision	Stable Diffusion 1.x, latent diffusion, UNet, DDPM, DDIM, Min-SNR, classifier-free guidance, EMA, VAE, GAN, DCGAN, PatchGAN, β -VAE, CycleGAN, AdaIN, spectral normalization, VGG and LPIPS perceptual loss.
Multimodal & Video	ViT, SigLIP, PaliGemma-style fusion, HRNet pose estimation, ST-GCN, CTR-GCN, graph convolution, NTU RGB+D 120.
Transformer Components	RoPE (incl. YaRN-scaled, pruned), GQA, SwiGLU, GeGLU, RMSNorm, multi-head attention, MLA, MQA, MTP, cross-attention.
Training & Infra	BF16, FP16, FP8, mixed precision, DDP, FSDP, NCCL, torch.compile, Flash-Attn 2, SDPA, gradient checkpointing, chunked cross-entropy, disk-backed token cache, atomic checkpointing, resumable pipelines.
Agentic ML	Multi-agent orchestration, provider-agnostic LLM routing, Ollama Cloud, vector and graph memory, Pydantic v2, pytest, Ruff.
Core	Python 3.12, PyTorch 2.x, Hugging Face, Diffusers, safetensors, SentencePiece, NumPy, W&B, Comet ML, img2dataset, Git, GitHub.
Hardware Validated	A100 80GB, RTX 5090 (Blackwell), RTX 6000 Ada, RTX 3090, P100, T4 (2×), RunPod, Kaggle.

WRITING

Multi-Head Latent Attention – A Technical Deep-Dive
643-line reference on KV-cache math, low-rank compression algebra, the absorption-trick

EDUCATION

B.Tech, Civil Engineering 2020 – 2024
Heritage Institute of Technology, Kolkata · heritageit.edu
Civil Engineering for the math, structures, and